

SECTION A*Answer all questions.*

1. (a) Describe an experiment that uses various absorbers to identify the type(s) of radiations emitted by a radioactive sample. Justify all steps in the experiment. [4]

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- (b) A radioactive sample of material has a half-life of 11.4 days and an initial activity of A_0 . Determine:

- (i) the decay constant; [2]

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(ii) the activity of the sample after 57.0 days in terms of A_0 ; [2]

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(iii) the **percentage decrease** in the number of nuclei in the sample after 57.0 days. [3]

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